KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
POLITEKNIK NEGERI MALANG
JURUSAN TEKNOLOGI INFORMASI
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Mata Kuliah	:	Pemograman Framework
Program Studi	:	D4 – Teknik Informatika
Semester	:	6

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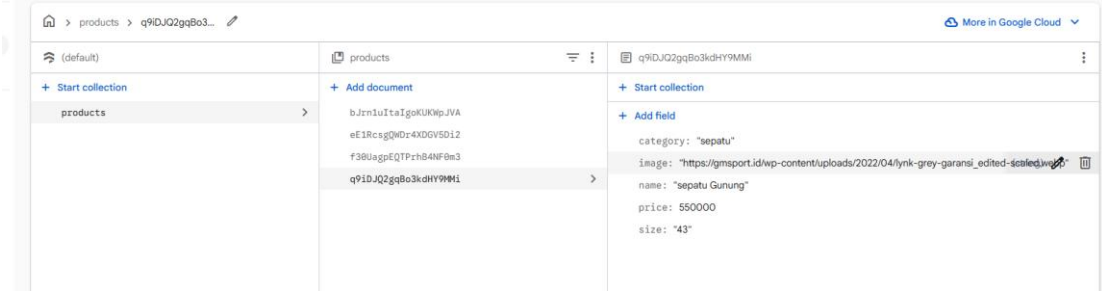
Jobsheet ke-	:	8

Laporan Jobsheet

Praktikum 1

Langkah	Jawaban/Deskripsi
1	Siapkan project Next.js. Buat endpoint API /api/products. Pastikan data memiliki: - Id - Name - Category - Price - Image 
2	jalankan browser http://localhost:3000/api/produk

Praktikum Ke-2

Langkah	Jawaban/Deskripsi
1	Membuat file index.tsx pada folder views/products views \ produk [id].tsx index.tsx M

2	Modifikasi index.tsx
	src > views > produk > index.tsx > TampilProduk > products.map() callback
	You, now | 1 author (You) | Windsurf: Refactor | Explain
	`type ProductType = {`
	`  id: string;`
	`  name: string;`
	`  price: number;`
	`  size: string;`
	`  category: string;`
	`  image: string;`
	`};`
	Windsurf: Refactor | Explain | Generate JSDoc | X
	`const TampilProduk = ({ products }: { products: ProductType[] }) => {`
	`  return (`
	`  <div>`
	`  <h1>Daftar Produk</h1>`
	`  {products.map((product: ProductType) => (`
	`  <div key={product.id}>`
	`  <h2>{product.name}</h2>`
	`  <p>Ukuran: {product.size}</p>`
	`  <p>Harga: {product.price}</p>`
	`  <img src={product.image} alt={product.name} />`
	`  <p>Kategori: {product.category}</p>`
	`  </div>` You, now • Uncommitted changes
	`  ))}`
	`  </div>`
	`  );`
	`};`
	`export default TampilProduk;`

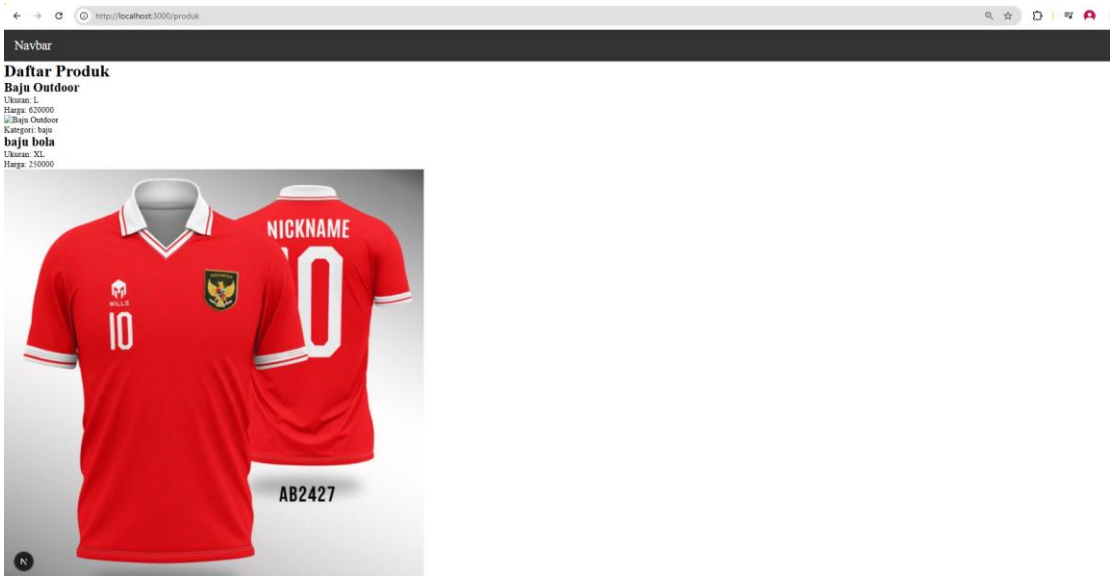
3	Buka dan modifikasi file index.tsx pada pages/produk/

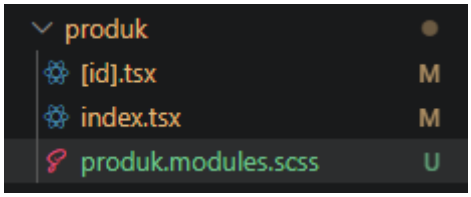
src > pages > produk > index.tsx > ...

You, 22 seconds ago | 1 author (You)

```
import { useRouter } from "next/router";
import { useEffect, useState } from "react";
import TampilanProduk from "../../views/product";      You, 21 seconds ago

Windsurf: Refactor | Explain | Generate JSDoc | X
const kategori = () => {

    const [products, setProducts] = useState([]);


    useEffect(() => {
        fetch("/api/produk")
            .then((response) => response.json())
            .then((responsedata) => {
                setProducts(responsedata.data);
                // console.log("Data produk:", responsedata.data);
            })
            .catch((error) => {
                console.error("Error fetching produk:", error);
            });
    }, []);

    return (
        <div>
            <TampilanProduk products={products} />
        </div>
    );
};

export default kategori;
```

4 Jalankan browser http://localhost:3000/produk



5	Pada folder produk buat file produk.modules.scss dan modifikasi 
6	Modifikasi Pada file index.tsx pada folder pages/views/product

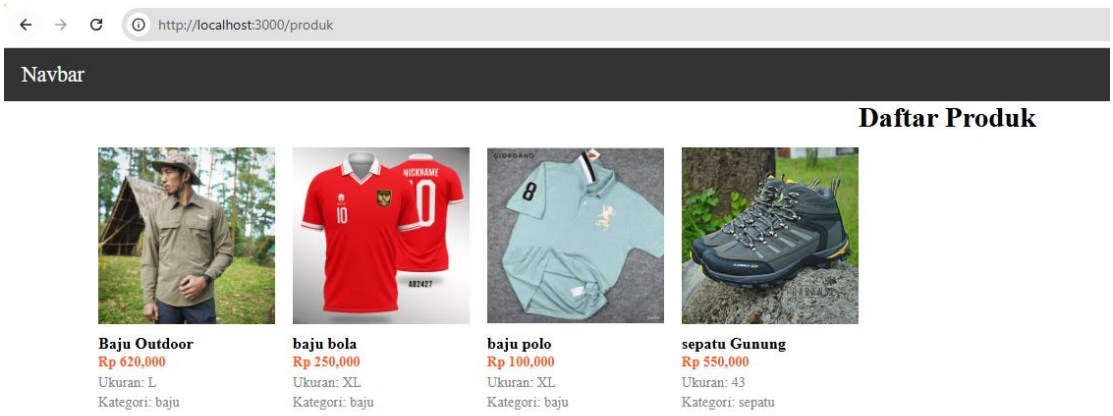
src > pages > produk > product.module.scss > .produk > &_content > &_item > &_name

```scss
.produk {
    width: 100%;
    padding: 0 5% ;
    &_title {
        text-align: center;
        font-size: 32px;
        font-weight: bold;
        margin-bottom: 16px;
    }

    &_content {
        display: flex;
        flex-wrap: wrap;
        gap: 20px;
        &_item {
            &_image {
                width: 100%;
                height: 200px;
                object-fit: cover;
                border-radius: 4px ;
                margin-bottom: 12px;            }
            &_name {
                font-size: 18px;
                font-weight: 600;
                margin-top: 8px;
            }
            &_price {
                font-size: 16px;
                color: #ff5722;
                font-weight: bold;
            }
             &_size {
                color: #888;
                margin-top: 4px;
            }
             &_category {
                color: #888;
                margin-top: 4px;
        }
    }
    }
```

src > pages > views > product > index.tsx > TampilanProduk > products.map() callback

```
import styles from "../../produk/product.module.scss";

Windsurf: Refactor | Explain
type ProductType = {
    id: string;
    name: string;
    price: number;
    size: string;
    category: string;
    image: string;
};

Windsurf: Refactor | Explain | Generate JSDoc | X
const TampilanProduk = ({ products }: { products: ProductType[] }) => {
    return (
        <div className={styles.produk}>
            <h1 className={styles.produk_title}>Daftar Produk</h1>
            <div className={styles.produk_content}>
            {products.map((product: ProductType) => (
                <div key={product.id} className={styles.produk_content_item}>
                    <div className={styles.produk_content_item_image}>
                    <img src={product.image} alt={product.name} width={200} />
                    </div>
                    <h4 className={styles.produk_content_item_name}>{product.name}</h4>
                    <p className={styles.produk_content_item_price}>Rp {product.price.toLocaleString()}</p>
                    <p className={styles.produk_content_item_size}>Ukuran: {product.size}</p>
                    <p className={styles.produk_content_item_category}>Kategori: {product.category}</p>
                </div>
            ))}
            </div>
        </div>
    );
};

export default TampilanProduk;
```

7 Jalankan Browser



Praktikum Ke-3

Langkah	Jawaban/Deskripsi
1	Modfikasi file index.tsx pada folder views/product/index.tsx

src > pages > views > product > index.tsx > TampilanProduk

```tsx
import styles from "../../produk/product.module.scss";

Windsurf: Refactor | Explain
type ProductType = {
    id: string;
    name: string;
    price: number;
    size: string;
    category: string;
    image: string;
};

Windsurf: Refactor | Explain | Generate JSDoc | X
const TampilanProduk = ({ products }: { products: ProductType[] }) => {
    return (
        <div className={styles.produk}>
            <h1 className={styles.produk_title}>Daftar Produk</h1>
            <div className={styles.produk_content}>
            {/* {products.map((product: ProductType) => (
                <div key={product.id} className={styles.produk_content_item}>
                    <div className={styles.produk_content_item_image}>
                    <img src={product.image} alt={product.name} width={200} />
                    </div>
                    <h4 className={styles.produk_content_item_name}>{product.name}</h4>
                    <p className={styles.produk_content_item_price}>Rp {product.price.toLocaleString()}</p>
                    <p className={styles.produk_content_item_size}>Ukuran: {product.size}</p>
                    <p className={styles.produk_content_item_category}>Kategori: {product.category}</p>
                </div>
            ))} */}

            <div className={styles.produk_content_skeleton}>
                <div className={styles.produk_content_skeleton_image}></div>
                <div className={styles.produk_content_skeleton_name}></div>
                <p className={styles.produk_content_skeleton_price}></p>
                <p className={styles.produk_content_skeleton_size}></p>
                <p className={styles.produk_content_skeleton_category}></p>
            </div>
            </div>
        </div>
    );
};
```

Modifikasi file product.module.scss

src > pages > produk > product.module.scss > @keyframes identifier

```scss
.produk {
    width: 100%;
    padding: 0 5% ;
    &_title {
        text-align: center;
        font-size: 32px;
        font-weight: bold;
        margin-bottom: 16px;
    }

    &_content {
        display: flex;
        flex-wrap: wrap;
        gap: 20px;
        &_skeleton {
            width: 200px;
            padding: 16px;
            border: 1px solid #eee;
            border-radius: 8px;
            box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
            display: flex;
            flex-direction: column;
            align-items: center;
            animation: identifier 1.5s infinite ease-in-out;
            &_image {
                width: 100%;
                height: 200px;
                background-color: #e0e0e0;
                border-radius: 4px ;
                margin-bottom: 12px;            }
            &_name {
                width: 80%;
                height: 20px;
                background-color: #e0e0e0;
                border-radius: 4px;
                margin-bottom: 8px;
            }
            &_price {
                width: 40%;
                height: 18px;
                background-color: #e0e0e0;
```

```
43                  }
44                   &_size {
45                     width: 30%;
46                     height: 16px;
47                     background-color: #e0e0e0;
48                     border-radius: 4px;
49                     margin-bottom: 8px;
50                  }
51                   &_category {
52                     width: 60%;
53                     height: 16px;
54                     background-color: #e0e0e0;
55                     border-radius: 4px;
56                     margin-bottom: 8px;
57              }
58          }
59
60              &_item {
61                  &_image {
62                      width: 100%;
63                      height: 200px;
64                      object-fit: cover;
65                      border-radius: 4px;
66                      margin-bottom: 12px;
67                  }
68                  &_name {
69                      font-size: 18px;
70                      font-weight: 600;
71                      margin-bottom: 8px;
72                  }
73                  &_price {
74                      font-size: 16px;
75                      color: #ff5722;
76                      font-weight: bold;
77                  }
```

<table>
<tr><td></td><td>

```
            &_size {
                font-size: 14px;
                color: #888;
                margin-bottom: 8px;
            }
            &_category {
                font-size: 14px;
                color: #888;
                margin-bottom: 8px;
            }
        }
    }
}

@keyframes identifier {
    0% {
        opacity: 1;
    }
    50% {
        opacity: 0;
    }
    100% {
        opacity: 1;
    }
}
```

</td></tr>
<tr><td>2</td><td>Jalankan browser maka akan muncul skeleton yang terdapat animasi berkedip

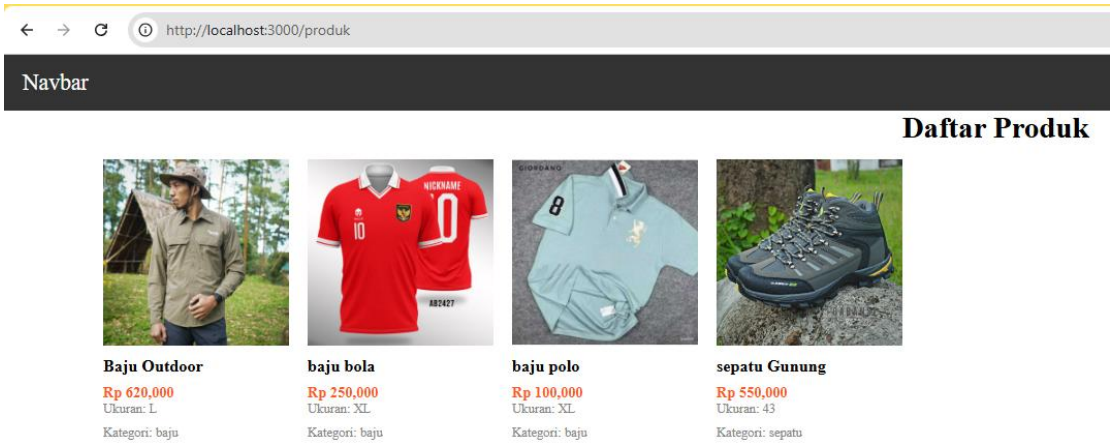
</td></tr>
<tr><td>3</td><td>Modifikasi pada index.tsx pada folder views/product/index.tsx</td></tr>
</table>

```
Windsurf: Refactor | Explain | Generate JSDoc | X
const TampilanProduk = ({ products }: { products: ProductType[] }) => {
    return (
        <div className={styles.produk}>
            <h1 className={styles.produk_title}>Daftar Produk</h1>
            <div className={styles.produk_content}>
                {products.length > 0 ? (
                    products.map((product: ProductType) => (
                        <div key={product.id} className={styles.produk_content_item}>
                            <div className={styles.produk_content_item_image}>
                                <img src={product.image} alt={product.name} width={200} />
                            </div>
                            <h4 className={styles.produk_content_item_name}>{product.name}</h4>
                            <p className={styles.produk_content_item_price}>Rp {product.price.toLocaleString()}</p>
                            <p className={styles.produk_content_item_size}>Ukuran: {product.size}</p>
                            <p className={styles.produk_content_item_category}>Kategori: {product.category}</p>
                        </div>
                    ))
                ) : (
                    <p>Tidak ada produk yang tersedia.</p>
                )}

            <div className={styles.produk_content_skeleton}>
                <div className={styles.produk_content_skeleton_image}></div>
                <div className={styles.produk_content_skeleton_name}></div>
                <p className={styles.produk_content_skeleton_price}></p>
                <p className={styles.produk_content_skeleton_size}></p>
                <p className={styles.produk_content_skeleton_category}></p>
            </div>
            </div>
        </div>
    );
};
```

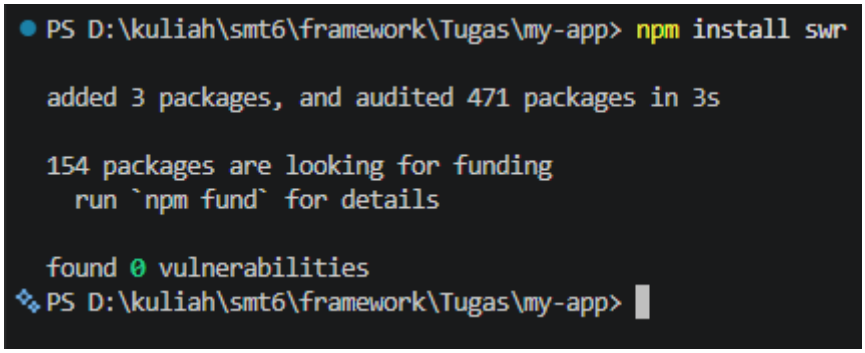
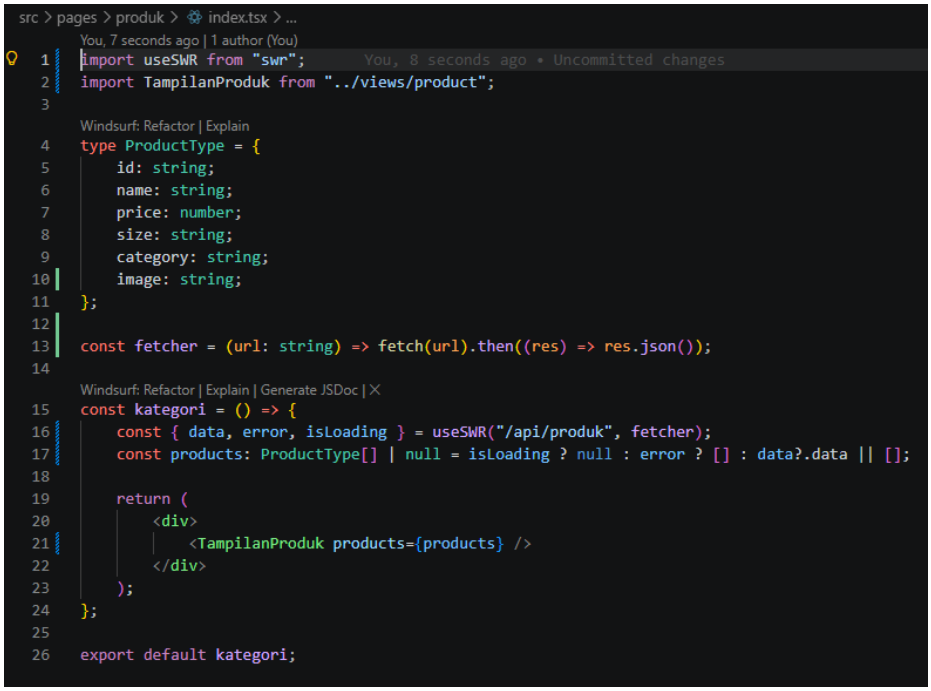
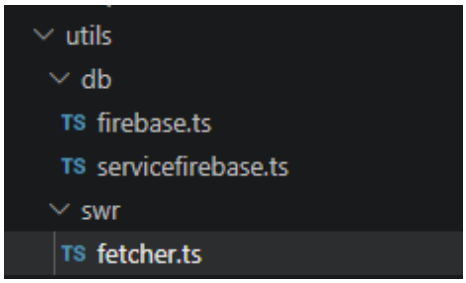
4

Jalankan browser

- Jika dijalankan akan muncul skeletonnya terlebih dahulu setelah itu muncul gambar dan informasinya



Praktikum Ke-4

Langkah	Jawaban/Deskripsi
1	Install SWR 
2	Buka dan modifkasi file index.tsx pada folder pages/product/ 
3	Buat folder swr pada utils dan tambahkan file dengan nama fetcher.js 

Modifikasi file fetcher.ts

```
src > pages > utils > swr > TS fetcher.ts > ...
1  const fetcher = (url: string) => fetch(url).then((res) => res.json());
2
3  export default fetcher;
4
```

Modifikasi file index.tsx pada folder pages/produk

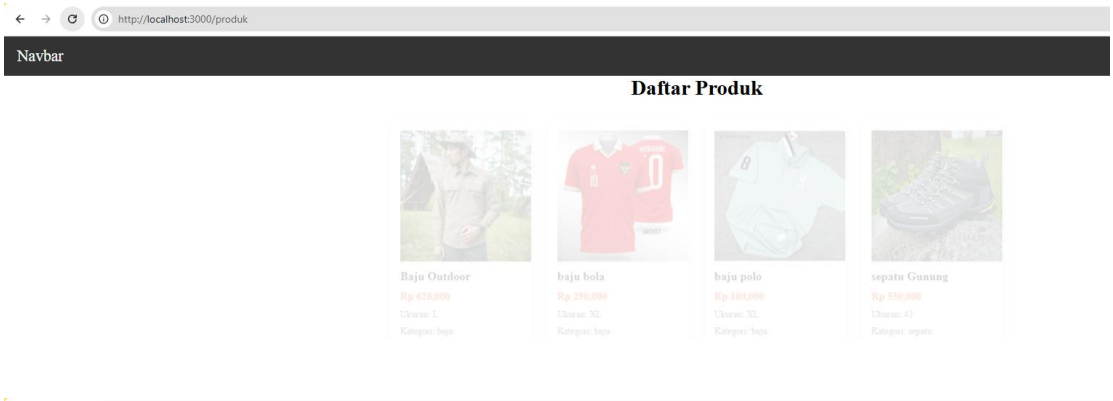
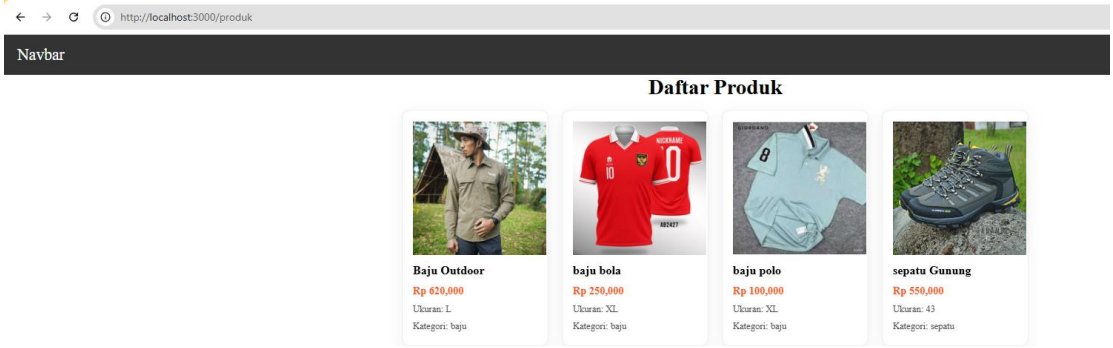
```
src > pages > produk > index.tsx > ...
You, 21 seconds ago | 1 author (You)
1  import useSWR from "swr";
2  import TampilanProduk from "../views/product";
3  import fetcher from "../utils/swr/fetcher";
4
Windsurf: Refactor | Explain
5  type ProductType = {
6      id: string;
7      name: string;
8      price: number;
9      size: string;
10     category: string;
11     image: string;
12 };
13         You, 1 hour ago • JS 7
Windsurf: Refactor | Explain | Generate JSDoc | X
14 const kategori = () => {
15     const { data, error, isLoading } = useSWR("/api/produk", fetcher);
16     const products: ProductType[] | null = isLoading ? null : error ? [] : data?.data || [];
17
18     return (
19         <div>
20             <TampilanProduk products={products} />
21         </div>
22     );
23 };
24
25 export default kategori;
```

Tugas 1

Langkah	Jawaban/Deskripsi
1	Jelaskan perbedaan: - Client Side Rendering - Server Side Rendering - Static Site Generation Jawab : - Client Side Rendering (CSR): Browser (HP/Laptop) yang bekerja merender tampilan menggunakan JavaScript setelah menerima data dari server.

	Contohnya adalah penggunaan useEffect untuk mengambil data API di sisi frontend. - Server Side Rendering (SSR): Server yang bekerja menyusun halaman menjadi HTML utuh setiap kali ada permintaan, sehingga saat sampai di browser, halaman sudah siap tampil. - Static Site Generation (SSG): Halaman dibuat menjadi file statis sekali saja saat proses build. Karena sudah jadi, file ini sangat cepat saat diakses pengguna karena tidak perlu diproses ulang

Tugas 2

Langkah	Jawaban/Deskripsi
1	Buat halaman produk dengan: - Skeleton loading - Animasi  

Tugas 3

Langkah	Jawaban/Deskripsi
1	Refactor kode dari useEffect menjadi SWR.

src > pages > produk > index.tsx > kategori

You, 1 minute ago | 1 author (You)

```
1  import useSWR from "swr";
2  import TampilanProduk from "../views/product";
3  import fetcher from "../utils/swr/fetcher";
4
   Windsurf: Refactor | Explain
5  type ProductType = {
6      id: string;
7      name: string;
8      price: number;
9      size: string;
10     category: string;
11     image: string;
12 };
13
   Windsurf: Refactor | Explain | Generate JSDoc | X
14 const kategori = () => {
15     const { data, error } = useSWR("/api/produk", fetcher);      You, now • Uncommitted
16     const products: ProductType[] | null = error ? [] : data?.data || null;
17
18     return (
19         <div>
20             <TampilanProduk products={products} />
21         </div>
22     );
23 };
24
25 export default kategori;
```